

SIMATS ENGINEERING

CO3- ASSESSMENT TOOL 3 ERROR ANALYSIS TASK

COURSE NAME: DIGITAL SIGNAL PROCESSING COURSE CODE: ECA09

COURSE OUTCOME COVERED

CO 3: Evaluate finite word-length effects and multirate signal processing techniques, including decimation, interpolation, and polyphase decomposition. **Bloom Level mapped- BL3,BL4,BL5**

Topics Covered:

Quantization- Truncation and rounding errors - Quantization noise - Limit cycle oscillations – Signal scaling. Parametric and Non-parametric Spectral estimation. Decimation-Interpolation-Sampling rate conversion-Implementation of sampling rate conversion-Polyphase Decomposition -5G Decimation/Interpolation: For mmWave beamforming

Assessment Tool number	: CO3-AT 3
Assessment Tool Weightage	: 34%
Assessment Tool Name	: Error Analysis Task
Duration of this assessment	: 60 Minutes
Marks	: 25 Marks
Type of Assessment Conduction	: Hybrid

QUESTION:

1. An IIR filter implemented with 6-bit arithmetic gives the output sequence: 0.50, 0.25, 0.12, 0.06, 0.03, 0.03, 0.03, 0.03.... Debug the reason for the constant output. (10) (BL5) (WK4)

Code of Conduct:

I Shreeram BS / 192412303 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question No	Error Identification 20%	Root Cause Analysis 30%	Correction 25%	Application of Concepts 15%	Clarity of Presentation 10%

Debugging an IIR Filter with Constant Output (10 Marks)

Given Data

An IIR filter implemented with 6-bit fixed-point arithmetic produces the following output sequence:

0.50, 0.25, 0.12, 0.06, 0.03, 0.03, 0.03, ...

Objective

Identify the implementation error(s), analyze the root cause, and provide the corrected solution with proper DSP justification.

Errors Identified

1. Finite Word Length (FWL) Error

The filter uses 6-bit arithmetic, which provides limited precision. Very small values cannot be represented accurately, resulting in a loss of numerical accuracy.

Effect: Reduced precision in filter computations.

2. Quantization Error

Each arithmetic operation (addition and multiplication) is quantized due to the limited number of bits.

For example,

- Ideal next output:
 $0.03 \times 0.5 = 0.015$
- Since 0.015 cannot be represented accurately using 6-bit arithmetic, it is quantized back to 0.03.

Effect: The output stops decreasing.

3. Rounding/Truncation Error

If truncation (or low-precision rounding) is used after each arithmetic operation, additional errors are introduced. These errors accumulate because an IIR filter contains a feedback loop.

Effect: Error accumulation in the recursive computation.

4. Zero-Input Limit Cycle Oscillation

The recursive feedback repeatedly feeds the quantized value back into the filter. As a result, instead of converging to zero, the output remains constant.

Effect: The filter becomes stuck at a constant non-zero value.

Root Cause Analysis

Ideal Output	Actual Output	Reason
0.50	0.50	Correct
0.25	0.25	Correct
0.125	0.12	Quantization Error
0.0625	0.06	Quantization Error
0.03125	0.03	Quantization Error
0.015625	0.03	Cannot be represented in 6-bit arithmetic
0.0078125	0.03	Zero-input limit cycle

Root Cause

Initially, the output decays normally. However, after reaching 0.03, the next expected value (≈ 0.015) is smaller than the resolution of the 6-bit fixed-point representation. Due to quantization, this value is rounded or truncated back to 0.03. Because the filter is recursive, this quantized value is continuously fed back, producing a constant output, known as a zero-input limit cycle oscillation.

Proposed Corrections

1. Increase the arithmetic precision (12-bit, 16-bit, or floating-point arithmetic).
2. Use rounding instead of truncation to reduce quantization error.
3. Properly scale the filter coefficients and internal variables.
4. Use higher-precision accumulation in the feedback path to minimize recursive quantization errors.

DSP Justification

IIR filters are highly sensitive to finite-word-length effects because their outputs are recursively fed back into the system. Quantization errors introduced in one iteration become inputs for subsequent iterations, causing error accumulation. When the arithmetic precision is too low, these errors prevent the output from converging to zero and produce a zero-input limit cycle, resulting in a constant non-zero output.

Conclusion

The constant output 0.03 is caused by finite-word-length effects in the 6-bit fixed-point implementation. The main implementation errors are finite word length, quantization error, rounding/truncation error, and zero-input limit cycle oscillation. Increasing arithmetic precision, using rounding instead of truncation, and improving coefficient scaling effectively eliminate the problem and restore the expected filter response.